

Edexcel Single Chemistry (1CI0) from 2016 Topic C1a&b				
Topic	Student Checklist	R	A	G
Topic 1a – Key concepts in chemistry Edexcel Single Chemistry (1CI0) from 2016 Topic C1a&b	Describe how the Dalton model of an atom has changed over time because of the discovery of subatomic particles			
	Describe the structure of an atom as a nucleus containing protons and neutrons, surrounded by electrons in shells			
	Recall the relative charge and relative mass of: a proton, a neutron and an electron			
	Explain why atoms contain equal numbers of protons and electrons			
	Describe the nucleus of an atom as very small compared to the overall size of the atom			
	Recall that most of the mass of an atom is concentrated in the nucleus			
	Recall the meaning of the term mass number of an atom			
	Describe atoms of a given element as having the same number of protons in the nucleus and that this number is unique			
	Describe what isotopes are			
	Calculate the numbers of protons, neutrons and electrons in atoms given the atomic number and mass number			
	Explain how the existence of isotopes results in relative atomic masses of some elements not being whole numbers			
	HT ONLY: Calculate the relative atomic mass of an element from the relative masses and abundances of its isotopes			
	Describe how Mendeleev arranged the elements known at that time, in a periodic table by using properties of these elements and their compounds			
	Describe how Mendeleev used his table to predict the existence and properties of some elements not discovered by then			
	Explain that Mendeleev thought he had arranged elements in order of increasing relative atomic mass but this was not always true			
	Explain the meaning of atomic number of an element in terms of position in the periodic table and number of protons in the nucleus			
	Describe how elements are arranged in the groups and periods of the periodic table			
	Identify elements as metals or non-metals according to their position in the periodic table, explaining this division in terms of atomic structure			
	Predict the electronic configurations of the first 20 elements in the periodic table as diagrams and in the form 2.8.1 etc			
	Explain how the electronic configuration of an element is related to its position in the periodic table			
	Explain how ionic bonds are formed to produce cations and anions, including the use of dot and cross diagrams			
	Recall that an ion is an atom or group of atoms with a positive or negative charge			
	Calculate the numbers of protons, neutrons and electrons in simple ions given the atomic number and mass number			
	Explain the formation of ions in ionic compounds from their atoms, limited to compounds of elements in groups 1, 2, 6 and 7			
	Explain the use of the endings –ide and –ate in the names of compounds			
	Deduce the formulae of ionic compounds given the formulae of the constituent ions			
	Explain the structure of an ionic compound including a description of the lattice and electrostatic forces			

Topic 1b – Key concepts in chemistry	Explain how a covalent bond is formed when a pair of electrons is shared between two atoms			
	Recall that covalent bonding results in the formation of molecules			
	Recall the typical size (order of magnitude) of atoms and small molecules			
	Explain the formation of simple molecular, covalent substances, using dot and cross diagrams, including: H, HCl, H ₂ O, CH ₄ , O ₂ , CO ₂			
	Explain why elements and compounds can be classified as: ionic, simple molecular (covalent), giant covalent and metallic			
	Explain how the structure and bonding of substances results in different physical properties			
	Explain the properties of ionic compounds limited to: melting/boiling points, forces between ions and conductivity			
	Explain the properties of typical covalent, simple molecular compounds limited to: melting/boiling points, forces between ions and conductivity			
	Recall that graphite and diamond are different forms of carbon and that they are examples of giant covalent substances			
	Describe the structures of graphite and diamond			
	Explain, in terms of structure and bonding, why graphite and diamond have different uses			
	Explain the properties of fullerenes including C ₆₀ and graphene in terms of their structures and bonding			
	Describe, using poly(ethene) as the example, that simple polymers consist of large molecules containing chains of carbon atoms			
	Explain the properties of metals, including malleability and the ability to conduct electricity			
	Describe the limitations of particular representations and models, to include dot & cross, ball & stick models & 2/3D			
	Describe the properties of most metals			
	Calculate relative formula mass given relative atomic masses			
	Calculate the formulae of simple compounds from reacting masses and understand that these are empirical formulae			
	Deduce: empirical formula of a compound from the formula of its molecule			
	Deduce: molecular formula of a compound from its empirical formula and its relative molecular mass			
	Describe an experiment to determine the empirical formula of a simple compound such as magnesium oxide			
	Explain the law of conservation of mass applied to: a closed system and a non-enclosed system			
	Calculate masses of reactants and products from balanced equations, given the mass of one substance			
	Calculate the concentration of solutions in g dm ⁻³			
	HT ONLY: Recall what one mole of particles of a substance is defined as			
	HT ONLY: Calculate the number of: moles of particles of a substance in a given mass of that substance and vice versa			
	HT ONLY: Calculate the number of: particles of a substance in a given number of moles of that substance and vice versa			
	HT ONLY: Calculate the number of: particles of a substance in a given mass of that substance and vice versa			
	HT ONLY: Explain why, in a reaction, the mass of product formed is controlled by the mass of the reactant which is not in excess			
	HT ONLY: Deduce the stoichiometry of a reaction from the masses of the reactants and products			

Edexcel Single Chemistry (1CI0) from 2016 Topics C6&7				
Topic	Student Checklist	R	A	G
Topic 6 – Groups in the periodic table	Explain why some elements can be classified as alkali metals, halogens or noble gases, based on their position in the periodic table			
	Recall the physical properties of alkali metals			
	Describe the reactions of lithium, sodium and potassium with water			
	Describe the pattern in reactivity of the alkali metals, lithium, sodium and potassium, with water; and use this pattern to predict the reactivity of other alkali metals			
	Explain this pattern in reactivity in terms of electronic configurations			
	Recall the colours and physical states of chlorine, bromine and iodine at room temperature			
	Describe the pattern in the physical properties of the halogens, chlorine, bromine and iodine, and use this pattern to predict the physical properties of other halogens			
	Describe the chemical test for chlorine			
	Describe the reactions of the halogens, chlorine, bromine and iodine, with metals to form metal halides, and use this pattern to predict the reactions of other halogens			
	Recall that the halogens, chlorine, bromine and iodine, form hydrogen halides which dissolve in water to form acidic solutions, and use this pattern to predict the reactions of other halogens			
	Describe the relative reactivity of the halogens chlorine, bromine and iodine, as shown by their displacement reactions with halide ions and use this to predict the reactions of astatine			
	HT ONLY: Explain why these displacement reactions are redox reactions in terms of gain and loss of electrons, identifying which of these are oxidised and which are reduced			
	Explain the relative reactivity of the halogens in terms of electronic configurations			
	Explain why the noble gases are chemically inert, compared with the other elements, in terms of their electronic configurations			
	Explain how the uses of noble gases depend on their inertness, low density and/or non-flammability			
	Describe the pattern in the physical properties of some noble gases and use this pattern to predict the physical properties of other noble gases			
Topic 7 - Rates of reaction and energy changes	<i>Core Practical: Investigate the effects of changing the conditions of a reaction on the rates of chemical reactions by: measuring the production of a gas/observing a colour change</i>			
	Suggest practical methods for determining the rate of a given reaction			
	Explain how reactions occur by discussing the collision theory			
	Explain the effects on rates of reaction of changes in temperature, concentration, surface area to volume ratio and pressure in terms of frequency and energy of collisions			
	Interpret graphs of mass, volume or concentration of reactant or product against time			
	Describe what a catalyst is			
	Explain how the addition of a catalyst increases the rate of a reaction in terms of activation energy			
	Recall that enzymes are biological catalysts and that enzymes are used in the production of alcoholic drinks			
	Recall when chemical changes occur that they cause changes in heat energy			
	Describe the differences between endothermic and exothermic in terms of energy taken in or given out			
	Recall if bonds are broken or made for each of the following reactions: endothermic and exothermic			
	Describe why the overall heat energy change for a reaction is exothermic or endothermic in terms of bonds being made or broken			
	HT ONLY: Calculate the energy change in a reaction given the energies of bonds (in kJ mol⁻¹)			
	Explain the term activation energy			
	Draw and label reaction profiles for endothermic and exothermic reactions, identifying activation energy			

Edexcel Single Chemistry (1CI0) from 2016 Topic C8				
Topic	Student Checklist	R	A	G
Topic 8 – Fuels and Earth science	Recall what a hydrocarbon is			
	Describe and explain what crude oil is and why it is important			
	Describe and explain the separation of crude oil into simpler, more useful mixtures by the process of fractional distillation			
	Recall the names and uses of the following fractions: gases, petrol, kerosene, diesel oil, fuel oil and bitumen			
	Explain how hydrocarbons in different fractions differ from each other in terms of boiling point, number of C & H's, flammability and viscosity			
	Explain what a homologous series of hydrocarbon compounds is			
	Describe the complete combustion of hydrocarbon fuels including energy changes and products			
	Explain why the incomplete combustion of hydrocarbons can produce carbon and carbon monoxide			
	Explain how carbon monoxide behaves as a toxic gas			
	Describe the problems caused by incomplete combustion in appliances that use carbon compounds as fuels			
	Explain how impurities in some hydrocarbon fuels result in the production of sulfur dioxide			
	Explain some problems associated with acid rain			
	Explain why, when fuels are burned in engines, oxides of nitrogen are formed and that they are pollutants			
	Evaluate the advantages and disadvantages of using hydrogen, rather than petrol, as a fuel in cars			
	Recall the names and sources of some renewable fossil fuels			
	Explain what cracking is and why it is necessary			
	Recall that the gases produced by volcanic activity formed the Earth's early atmosphere			
	Describe what the Earth's early atmosphere was thought to contain			
	Explain what the oceans were formed from			
	Explain why the amount of carbon dioxide in the atmosphere decreases when the oceans were formed			
	Explain how the growth of primitive plants changes the composition of gases in the atmosphere			
	Describe the chemical test for oxygen			
	Describe and explain the greenhouse effect and name the gases that contribute to it			
	Evaluate the evidence for human activity causing climate change			
	Describe the potential effects on the climate of increased levels of carbon dioxide and methane generated by human activity			
	Describe how effects on the climate may be mitigated: consider scale, risk and environmental implications			

Edexcel Single Chemistry (1C10) from 2016 Topic C9a&b (Chem only)				
Topic	Student Checklist	R	A	G
Topic 9a – Separate chemistry 2 part a	Chem ONLY: Explain why the test for any ion must be unique			
	Chem ONLY: Describe flame tests to identify the following ions in solids: Li^+ , Na^+ , K^+ , Ca^{2+} , Cu^{2+} including the colours of the flames			
	Chem ONLY: Describe tests to identify the following ions: Al^{3+} , Ca^{2+} , Cu^{2+} , Fe^{2+} , Fe^{3+} , NH_4 using NaOH solution			
	Chem ONLY: Describe the chemical test for ammonia			
	Chem ONLY: Describe tests to identify the following ions: CO_3^{2-} , SO_4^{2-} , Cl^- , Br^- , I^-			
	<i>Chem ONLY: Core Practical: Identify the ions in unknown salts, using the tests for the specified cations and anions in the specification</i>			
	Chem ONLY: Identify the ions in unknown salts, using results of the tests stated			
	Chem ONLY: Describe that instrumental methods of analysis are available and that these may improve sensitivity, accuracy and speed of tests			
	Chem ONLY: Evaluate data from a flame photometer to determine the concentration of ions in dilute solution using a calibration curve			
	Chem ONLY: Evaluate data from a flame photometer to identify metal ions by comparing the data with reference data			
	Chem ONLY: To identify metal ions by comparing the data with reference data			
	Chem ONLY: Recall the formulae of molecules of the alkanes, methane, ethane, propane and butane, and draw the structure of these			
	Chem ONLY: Explain why the alkanes are saturated hydrocarbons			
	Chem ONLY: Recall the formulae of molecules of the alkenes, ethene, propene, butene, and draw the structures (but-1-ene and but-2-ene only)			
	Chem ONLY: Explain why the alkenes are unsaturated hydrocarbons			
	Chem ONLY: Recall the addition reaction of ethene with bromine, showing the structures of reactants and products, and extend this to other alkenes			
	Chem ONLY: Explain how bromine water is used to distinguish between alkanes and alkenes			
	Chem ONLY: Describe how the complete combustion of alkanes and alkenes involves the oxidation of the hydrocarbons, name the products			

Topic 9b – Separate chemistry 2 part b	Chem ONLY: Recall that a polymer is a substance of high average relative molecular mass made up of small repeating units			
	Chem ONLY: Describe how ethene molecules can combine together in a polymerisation reaction			
	Chem ONLY: Describe that the addition polymer formed is called polyethene			
	Chem ONLY: Describe how other addition polymers can be made by combining together other monomer molecules containing C=C			
	Chem ONLY: Describe how to deduce the structure of a monomer from the structure of an addition polymer and vice versa			
	Chem ONLY: Explain how the uses of polymers are related to their properties and vice versa			
	HT & Chem ONLY: Explain why polyesters are condensation polymers			
	HT & Chem ONLY: Explain how a polyester is formed when a monomer molecule containing two carboxylic acid groups is reacted with a monomer molecule containing two alcohol groups			
	HT & Chem ONLY: Explain how a molecule of water is formed each time an ester link is formed			
	Chem ONLY: Describe some problems associated with polymers including the availability of starting materials			
	Chem ONLY: Describe the advantages and disadvantages of recycling polymers, including economic implications, availability of starting materials and environmental impact			
	Chem ONLY: Recall that DNA is a polymer made from four different monomers called nucleotides			
	Chem ONLY: Recall that starch is a polymer based on sugars			
	Chem ONLY: Recall that proteins are polymers based on amino acids			
	Chem ONLY: Recall the formulae of molecules of the alcohols, methanol, ethanol, propanol and butanol, and draw the structures of these molecules, showing all covalent bonds			
	Chem ONLY: Recall that the functional group in alcohols is –OH			
	<i>Chem ONLY: Core Practical: Investigate the temperature rise produced in a known mass of water by the combustion of the alcohols ethanol, propanol, butanol and pentanol</i>			
	Chem ONLY: Recall the formulae of molecules of the carboxylic acids, methanoic, ethanoic, propanoic and butanoic acids, and draw the structures of these molecules, showing all covalent bonds			
	Chem ONLY: Recall that the functional group in carboxylic acids is –COOH			
	Chem ONLY: Recall that ethanol can be oxidised to produce ethanoic acid and extend this to other alcohols			
	Chem ONLY: Recall members of a given homologous series have similar reactions because their molecules contain the same functional group and use this to predict the products of other in series			
	Chem ONLY: Describe the production of ethanol by fermentation of carbohydrates in aqueous solution, using yeast to provide enzymes			
	Chem ONLY: Explain how to obtain a concentrated solution of ethanol by fractional distillation of the fermentation mixture			
	Chem ONLY: Compare the size of nanoparticles with the sizes of atoms and molecules			
	Chem ONLY: Describe how the properties of nanoparticulate materials are related to their uses including surface area to volume ratio of the particles they contain, including sunscreens			
	Chem ONLY: Explain the possible risks associated with some nanoparticulate materials			
	Chem ONLY: Compare, using data, the physical properties of glass and clay ceramics, polymers, composites and metals			
	Chem ONLY: Describe how the properties of nanoparticulate materials are related to their uses including surface area to volume ratio of the particles they contain, including sunscreens			
	Chem ONLY: Explain the possible risks associated with some nanoparticulate materials			
	Chem ONLY: Compare, using data, the physical properties of glass and clay ceramics, polymers, composites and metals			
	Chem ONLY: Explain why the properties of a material make it suitable for a given use and use data to select materials appropriate for specific uses			

